Zheguang Zhao

Senior Software Engineer
NetApp Inc.
Level 1, 550 Bourke Street
Melbourne, VIC 3000
Australia

Phone:	+61-433-440-582
Email:	zheguang.zhao@alumni.brown.edu
Homepage:	zheguang.github.io
LinkedIn:	www.linkedin.com/in/zheguang
Google Scholar:	goo.gl/DR8pSa

Education

Ph.D. in Computer Science, Brown University, Rhode Island, 2021
Advisors: Stan Zdonik, Seny Kamara
Readers: George Kollios, Moti Yung
Thesis: *Building a Structurally-Encrypted Relational Database*
Software: KafeDB

M.Sc. in Computer Science, Brown University, Rhode Island, 2016
Advisor: Stan Zdonik
Readers: Tim Kraska, Ugur Cetintemel
Thesis: *Approximate Data Structures for Visualization*

B.Sc. in Computer Science, University of Wisconsin at Madison, Wisconsin, 2012
Advisor: Jignesh Patel, Ben Liblit

Experiences

NetApp Inc., Cloud Storage, *Senior Software Engineer* (2024-Current)

Apache Kafka MCP Agent: agentic real-time analytics and performance tuning

Apache Kafka Connector for NetApp enterprise storage: TB-scale columnar data

ClickHouse query optimization: 5x speedup on text search using SIMD

Multi-cloud management platform: Automated OS upgrade for 5000+ nodes

The University of Melbourne, *Research Fellow in Secure Systems* (2023-Current)

Security on AI confidence attacks: model output manipulation via data perturbation [link]

Funded by U.S. Army Research Office, part of CATCH project

Hosted by Prof. Olya Ohrimenko and Prof. Ben Rubinstein

Technical University of Darmstadt, *Postdoc Researcher* (2021-2022)

Secure federated learning and databases with Intel SGX.

Hosted by Prof. Carsten Binnig

Aroki Systems (now part of MongoDB), *Database Scientist* (2017-2021)

KafeDB, an end-to-end encrypted relational database based on SparkSQL, [link]

State-of-the-art encrypted join: achieving optimal storage and time complexity

Los Alamos National Laboratory, *Research Intern* (2019)

AI model reconstruction of mixed dynamics in cyber-physical systems, with application to network verification and security

Hosted by Dr. Andrey Lokhov

Microsoft AI & Research, *Research Intern* (2017)

Constraint learning. Host: DMX Group

Intel Labs, *Research Intern* (2015)

Efficiency of machine learning algorithms in Apache Spark

In-memory transactional processing using non-volatile memory

Hadapt (now part of Teradata), *Software Engineer* (2013-2014)

Enterprise SQL-on-Hadoop system including query execution, storage engine, high availability and analytics

Kosmix (now part of @WalmartLabs), *Software Engineer* (2012-2013)

In-memory distributed queue system for in-house stream processing

Great Lakes Bioenergy Research Center, *Software Engineer* (2009-2012)

Scientific database for biological enzyme research

Open Source Software

Apache Kafka MCP Agent: automated real-time analytics and performance tuning [talk],[link]

Adapt system parameters for workloads

Query event streams by natural language

Apache Kafka Connector for AWS and NetApp Storage [talk],[PR]

Unlimited Parquet file size (Better than state-of-thart Confluent and Aiven)

Stable read over large S3 objects (e.g.,$O(TB)$)

Exactly-once transaction over unicode input

Asynchronous file discovery

AI tuning for optimal system parameters

ClickHouse query optimizations [link]

5x speedup on pattern matching [PR]

35% speed up on case-insensitive affix search [PR]

Paralle data ingestion with constant common table expression [PR]

KafeDB: End-to-End Structurally-Encrypted Relational Database [link]

SOTA encrypted join algorithm: achieving linear time and space complexity

Part of my PhD thesis

Approximate Computation using GPGPU [simulator][benchmark]

ML framework for Cypber-physical Systems [link]

Encrypted Searchable Signal [link]

Macau: statistical hypothesis testing based on resampling [link]

Machine learning algorithms in Spark [link]

Consistency control for machine learning algorithms [link]

R-tree in Rust [link]

Spark performance analysis tool [link]

VoltDB on non-volatile memory [link]

Publications

Benchmarking the Second Generation of Intel SGX Hardware
Proceedings of the 18th International Workshop on Data Management on New Hardware

Towards Decentralized Parameter Servers for Secure Federated Learning DATA, 2022

ACID-V: towards a New Class of DBMSs for Data Sharing
Polystores Workshop at VLDB, 2021

Encrypted Databases: from Theories to Systems
CIDR, January 2021

An Optimal Relational Database Encryption Scheme [link]
Cryptology ePrint Archive: Report 2020/274

Learning of Cyber-Physical Systems
Advanced Network Science Initiative, Los Alamos National Laboratory, 2019

Behavior of Large Random Graph. [link]
Randomized Algorithms for Counting, Integration and Optimization, Brown University, 2017

Signal Search.[link]
Brown University, 2017

Dynamic Query Refinement for Interactive Data Exploration
EDBT/ICDT Joint Conference, March 2020

Investigating the Effect of the Multiple Comparisons Problem in Visual Analysis
CHI Conference, April 2018

Controlling False Discoveries During Interactive Data Exploration
SIGMOD Conference, May 2017

Safe Visual Data Exploration
SIGMOD Conference, Demo, May 2017

Bridging the Gap between HPC and Big Data frameworks
VLDB Journal, 2017

Towards Sustainable Insights
CIDR Conference, January 2017

Towards a Benchmark for Interactive Data Exploration
IEEE Data Engineering Bulletin, 2016.

Larger-than-memory Data Management on Modern Storage Hardware for In-memory OLTP Database Systems
SIGMOD DaMoN Workshop, June 2016

VisTrees: Fast Indexes for Interactive Data Exploration
SIGMOD HILDA Workshop, June 2016

Data Tiering in Heterogeneous Memory Systems
EuroSys Conference, April 2016

Teaching

Security Analytics, The University of Melbourne, 2023.

Data Management Labs, TU Darmstadt, 2021.

Introduction to Database Systems, Brown University 2015, 2018

Advanced Topics in Database Systems, Brown University 2015

Supervision

Benedikt Völker, *Client-side Validation for Detecting the Model Poisoning Attack in Federated Learning*
M.Sc.'21, TU Darmstadt

Philipp Imporatori, *Building an Oblivious Relational Database*
M.Sc.'22, TU Darmstadt

Shan Li, *Gradient-based Attacks on Federated Learning*
M.Sc.'22, TU Darmstadt

Services

Reviewer: OSDI'16, TVCG'21, TODS'21, IEEE S&P (Journal)'24, IEEE S&P (Oakland)'24

Program committee: VLDB'22 (Demo Track)

Member: IACR, ACM

Honors

Graduate Fellowship, Brown University, 2014-2019

Dean's Honor List, University of Wisconsin at Madison, 2007, 2008, 2009

Eta Kappa Nu, 2009

Upsilon Pi Epsilon, 2010

Golden Key International Honour Society, 2010

Dissertation Fellowship, Brown University, 2018-2019

Funding

CATCH MURI by U.S. Army Resaerch Office, 2022

NSF Award #1514491, *III: Medium: 20/20: A System for Human-in-the-Loop Data Exploration*, 2015

NSF Award #1916335, *SBIR Phase I: Encrypted Databases: From Theory to Systems*, 2019

References

Available upon request.